

| Milestone Description                                                                                                              | Deliverable(s)                                                                                                  | Effort                | Timeline             |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|
| Establish unity project and setup with version control and team environment (GitHub repository, Unity Version Control, etc.)       | i. Private GitHub repository with development and main branch and a release template                            | Matt 100%             | 1/12/2026-1/19/2026  |
| Download dependencies (AR/XR toolkits, AI, etc.)                                                                                   | ii. Private Unity Repository                                                                                    | Cael 100%             |                      |
|                                                                                                                                    | i. Fully functioning Unity, C#, and Python editors                                                              | 50% Matt, 50% Cael    | 1/19/2026-1/26/2026  |
| Create Python development environment for backend development                                                                      | i. Fully functioning environment for both machines                                                              | 50% Matt, 50% Cael    | 1/26/2026-2/2/2026   |
|                                                                                                                                    | ii. Confirm GPU/CUDA availability (Matt 100%)                                                                   | 100% Matt             |                      |
| Draft Python FastAPI that encapsulates training and inference code                                                                 | i. Locally hosted FastAPI                                                                                       |                       |                      |
|                                                                                                                                    | ii. Training endpoint which trains model to be able to segment and classify relevant objects in scene           | 100% Matt             | 2/2/2026-3/2/2026    |
|                                                                                                                                    | iii. Inference endpoint that runs inference on trained model and returns locations of objects and their classes |                       |                      |
|                                                                                                                                    | iv. Testing of API endpoints' accessibility from Unity project scripts                                          |                       |                      |
| Develop UI by creating dynamic objects that represent user actions                                                                 | i. A fully functioning UI made in Unity Software (Cael 100%)                                                    | Cael 100%             | 2/2/2026-3/2/2026    |
| Research computer vision (CV) methodologies to discover what is optimal for our use case                                           | i. A deeper understanding of the methodologies required for the project's development                           | Matt 100%             | 3/2/2026-3/9/2026    |
| Experiment with different CV models for object detection (Dino, YOLO, etc.) to find one that works with given hardware constraints | i. A chosen CV model                                                                                            | Matt 100%             | 3/9/2026-3/23/2026   |
|                                                                                                                                    | ii. Qualitative results showing models' abilities                                                               |                       |                      |
| Implement chosen model                                                                                                             | i. Implement model into FastAPI such that it can be trained and inferenced using existing stub functions        | Matt 100%             | 3/23/2026-3/30/2026  |
| Implementation of calls to FastAPI by Unity project scripts                                                                        | i. Communication between front-end and back-end established and working                                         | Cael 100%             | 3/2/2026-3/17/2026   |
| Creation of interaction overlays in Unity that are displayed by the user                                                           | i. Fully implemented interactable overlays for the user to use                                                  | Cael 100%             | 2/2/2026-3/28/2026   |
| Integration testing to ensure individual components work together                                                                  | i. Fully functioning components that produce no errors when communicating                                       | 50% Matt, 50% Cael    | 3/9/2026-project end |
| Debugging of various systems                                                                                                       | i.No system errors in the final product                                                                         | Cael 50% and Matt 50% | 3/9/2026-project end |
| Deployment of application onto Meta Quest 3S head-mounted display                                                                  | i. A visible application ready for use on the AR device                                                         | Cael 50% and Matt 50% | 3/2/2026-project end |
| Presentation and demonstration of our final project                                                                                | i. A presentation delivered and the end of the project                                                          | Matt 50% and Cael 50% |                      |
|                                                                                                                                    | ii.Demonstration at CEAS expo                                                                                   | Matt 50% and Cael 50% | Project end          |